


National University of Computer and Emerging Sciences

MLOps & Production AI Engineering

# Building a Production-Ready MLOps AI Agent: Design, Develop, Deploy, and Automate

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**Build commands**

```
python3.12 -m venv .venv && .venv/bin/pip install -r requirements.txt
.venv/bin/pytest
.venv/bin/uvicorn agent.main:app --port 8001
scripts/start_mlflow.sh ; scripts/start_airflow.sh ; scripts/setup_cluster.sh
```

Toolchain: Python 3.12.3, MLflow 3.18.0, Airflow 3.4.0, DVC 3.67.1, kind 0.33.0 (Kubernetes 1.37), Docker 29

# 1. Scope and environment

This report describes the design, implementation and testing of an MLOps agent that answers operational questions by calling real tools on DVC, MLflow, Apache Airflow and Kubernetes. All work was done on my own machine against infrastructure I started myself. No third-party system was touched, and no attack or load testing of any kind was performed.

- **Host:** one Linux laptop (kernel 6.8, 12 cores, 7.4 GB RAM), Python 3.12.3.
- **Local services:**
  - MLflow 3.18.0 tracking server and model registry on `127.0.0.1:5000`.
  - Apache Airflow 3.4.0 (standalone) on `127.0.0.1:8080`, in its own virtual environment.
  - A kind cluster named `mlops` (Kubernetes 1.37) with one application namespace, `mlops`.
  - The agent API on `127.0.0.1:8001`. The DVC remote is a local folder.
- **Language model:** `openai/gpt-oss-120b` through Groq's OpenAI-compatible API. Only the user's question and the tool results (run ids, pod names, metrics) are sent to it.
- **Secrets:** API keys and the Airflow password live only in a git-ignored `.env` file. They are held as `SecretStr` and redacted from logs. No credential is in the repository.
- **Source:** `github.com/Mahad1090/MLOPs-Assignment-3`, developed on a feature branch and merged through pull request #1 after CI passed.

# 2. System architecture

## 2.1 Overview

A user sends a natural-language request to a FastAPI interface. The orchestrator gives the request, the conversation so far and the tool schemas to the language model. The model decides which tools to call. Each call passes through a guarded executor and the results go back to the model, which writes the final answer.

The model never supplies facts. Every identifier, metric, state and log line in an answer comes from a tool result, and the raw tool results are returned to the caller next to the answer.

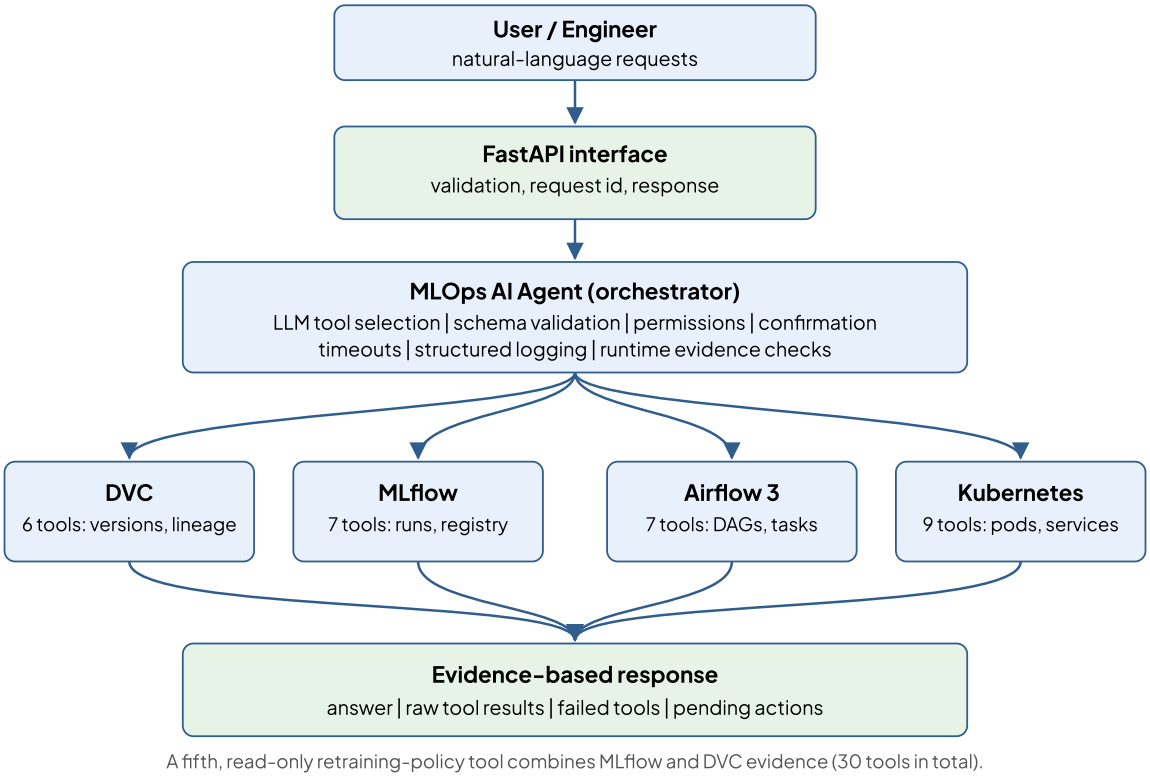



***Figure 1:*** *Logical architecture of the agent and its four tool integrations.*

## 2.2 Components

| Component | Responsibility |
|---|---|
| `agent/main.py` | FastAPI interface: `POST /chat`, `GET /tools`, `GET /health`. Pydantic validation rejects unknown fields and oversize input. |
| `agent/cli.py` | Terminal chat using the same orchestrator. Asks y/N before a pending action and can write transcripts. |
| `agent/orchestrator.py` | Agent loop with a 12-step budget, per-session history, the pending-action store and the runtime guards of section 7. |
| `agent/llm.py` | Client for an OpenAI-compatible chat API. Retries on rate limits and falls back to the next configured key. Tests replace it with a scripted model. |
| `agent/tools/base.py` | Tool registry and executor: validate, check permission, run with a time budget, log one structured line. |
| `agent/tools/*.py` | The 30 tools in five modules (DVC, MLflow, Airflow, Kubernetes, retraining policy). |
| `training/*.py` | Pipeline stages: validate, preprocess, train, evaluate, quality gate, register. |
| `dags/training_pipeline.py` | Airflow 3 DAG that runs those stages with retries and gating. |
| `api/main.py`, `kubernetes/` | The FastAPI prediction service and its Deployment, Service, ConfigMap and RBAC. |

## 2.3 Request flow

1. `POST /chat` is validated and given a request id.

2. The orchestrator sends the system prompt, earlier question and answer text and the tool schemas to the model.
3. For every tool call the model returns, the executor validates the arguments, runs the tool's permission check, then runs the tool in a worker thread with a time budget.
4. The outcome is one of seven statuses: `ok`, `error`, `unavailable`, `timeout`, `invalid_input`, `denied` or `confirmation_required`.
5. Tool results go back to the model as JSON. A result that is not `ok` carries no data, only the error and an instruction not to guess.
6. When the model stops calling tools, its text becomes the answer. The runtime then appends a notice for every failed tool and a list of actions awaiting confirmation, whatever the model wrote.

## 2.4 How provenance links the four systems

```
git tag data-v2.0 --> data/raw/dataset.csv.dvc (md5) --> DVC cache / remote (the bytes)
        |
        training/train.py hashes the file it actually reads and opens an MLflow run
        v
MLflow run tags: dvc.md5, dvc.dataset_version, dvc.pointer_commit, git.commit,
                 dvc.workspace_matches_pointer, airflow.dag_run_id, quality_gate
        v
MLflow model-version tags --> serving_model/model_meta.json (baked into the image)
        v
prediction-api  GET /version  reports model version, run id and dataset version
```

"Which dataset trained the model that is deployed?" is therefore answered by reading recorded values from end to end: Kubernetes `/version`, then the MLflow run, then the DVC md5, then the git tag.

## 2.5 Tool definitions and permissions

Every tool has a Pydantic input model that forbids unknown fields. `GET /tools` returns the full JSON schemas. The table below is generated from the live tool registry.

| Tool | What it does | Changes state |
|---|---|---|
| **dvc** | | |
| `dvc_list_tracked_datasets` | List datasets tracked by DVC (path, md5, size), pipeline outputs and remotes. | no |
| `dvc_current_version` | Current dataset version in the workspace: DVC md5, matching version tags, git commit, and whether the file on disk still matches the... | no |
| `dvc_compare_versions` | Compare a dataset between two git revisions (md5, size, rows, dvc diff). | no |
| `dvc_status` | Detect whether DVC-tracked data changed versus what is recorded, and whether the remote is in sync. | no |
| `dvc_find_version` | Resolve a dataset reference recorded for a training run (the dvc.md5 and/or git.commit tags of an MLflow run) to a dataset version tag... | no |

| | | |
|---|---|---|
| `dvc_reproduce_instructions` | Explain how to reproduce a previous dataset version, and verify it by fetching that version into a temporary directory and checking its md5. | no |
| **mlflow** | | |
| `mlflow_list_experiments` | List MLflow experiments with run counts and last activity. | no |
| `mlflow_list_runs` | List the most recent training runs of an experiment with metrics, parameters and the recorded dataset provenance (dvc.md5,... | no |
| `mlflow_get_run` | Full detail of one run by id or name: metrics, params, artifacts, quality gate and the DVC dataset reference recorded at training time. | no |
| `mlflow_best_run` | Rank recent runs by a metric and return the best one with its margin, quality gate and dataset provenance. | no |
| `mlflow_compare_runs` | Compare specific runs, or the last N runs, side by side. | no |
| `mlflow_compare_experiments` | Compare two or more experiments on a metric (run counts, best, mean). | no |
| `mlflow_model_registry` | Registered models, their versions, aliases, tags and source runs. | no |
| **airflow** | | |
| `airflow_list_dags` | List Airflow DAGs, whether they are paused and whether the agent may trigger them. | no |
| `airflow_list_dag_runs` | Most recent runs of a DAG with state, timing and the conf they ran with. | no |
| `airflow_get_dag_run` | Status of one DAG run (latest by default) with the state of every task. | no |
| `airflow_failed_tasks` | Identify the failed tasks of a DAG run (latest by default), separating the task that actually failed from downstream tasks that were... | no |
| `airflow_task_logs` | Execution details and the log tail of one task attempt. | no |
| `airflow_trigger_dag` | Start a run of an approved training DAG. | **yes, after confirmation** |
| `airflow_wait_for_dag_run` | Wait (up to a few minutes) for a DAG run to finish and return task states. | no |
| **kubernetes** | | |
| `k8s_list_workloads` | List Deployments and Pods of the ML application with readiness and restarts. | no |
| `k8s_pod_status` | Detailed status of one Pod: conditions, container states, restart reasons, probe configuration, resources and its events. | no |
| `k8s_pod_logs` | Tail the logs of a Pod (optionally of its previous, crashed container). | no |
| `k8s_services` | Inspect Services: type, ports, selector and which Pod endpoints are ready, i.e. | no |

| | | |
|---|---|---|
| `k8s_unhealthy_pods` | Find Pods that are not ready, container restarts and recent warning events. | no |
| `k8s_recent_events` | Recent Kubernetes events in the application namespace. | no |
| `k8s_deployed_version` | Report the deployed application version: image tag, version label, ConfigMap, rollout state and what the running app reports on /version... | no |
| `k8s_restart_deployment` | Rollout-restart the application Deployment. | **yes, after confirmation** |
| `k8s_scale_deployment` | Change the replica count of the application Deployment. | **yes, after confirmation** |
| **policy** | | |
| `evaluate_retraining_need` | Check a model's training run against the explicit retraining policy (model age, quality gate, dataset version drift, user-supplied... | no |

### Permission rules

- **Read tools** run immediately. Kubernetes reads are confined to the namespace `mlops`. DVC access is limited to an allowlist of read-only git and dvc sub-commands, run without a shell.
- **The three state-changing tools are never executed from a model tool call.** The call becomes a pending action. Only a later request that carries its `confirm_action_id` runs it. The id is single-use, bound to the session and expires after 10 minutes.
- `airflow_trigger_dag` accepts only DAG ids in `AIRFLOW_APPROVED_DAGS` and four typed override fields.
- `k8s_restart_deployment` and `k8s_scale_deployment` accept only the approved Deployment in the approved namespace, with at most 4 replicas.
- There is no tool to delete, exec, apply, read secrets, or write to DVC or git.
- `kubernetes/rbac.yaml` defines a matching least-privilege ServiceAccount and Role, so the agent can also run without an admin kubeconfig.

## 2.6 Design decisions

| Decision | Reason |
|---|---|
| DVC through `subprocess` with an allowlist | Easy to audit: a fixed list of `(binary, sub-command)` pairs, no shell, validated revisions. Old versions are fetched with `dvc get` into a temporary directory, so the workspace is never modified. |
| `MlflowClient` | Typed access to runs, metrics and the registry, as the assignment requires. Retries are reduced so an outage is reported in seconds. |
| Airflow REST API v2 with JWT | Airflow 3 removed `/api/v1` and basic auth. The agent logs in at `/auth/token` and refreshes the token once on HTTP 401. |
| Official Kubernetes Python client | Structured objects instead of parsing `kubectl` text, per-request timeouts and no shell. |
| Direct tool calling, no agent framework | The loop is short, so every safety property is in code that can be read and unit tested. |
| Separate Airflow virtual environment | Airflow pins hundreds of dependencies. The DAG runs each stage with the project interpreter, so ML libraries never have to satisfy those pins. |

| | |
|---|---|
| Model baked into the image | The Pods do not depend on the MLflow server at run time. `model_meta.json` keeps the link back to the run and dataset. |
| kind | One Docker container, the lowest memory footprint for a 2–replica deployment on a laptop. |

# 3. Task 1: DVC data management

## 3.1 Dataset versions

The data is the Wisconsin diagnostic breast-cancer set from scikit-learn, written by `scripts/make_dataset.py`. Two versions are tracked with DVC and tagged in git, so provenance questions have more than one possible answer.

| Git tag | Rows | Size | DVC md5 |
|---|---|---|---|
| `data-v1.0` | 400 | 85,539 B | `ac465752f88767fa7b7a2a45b3338f81` |
| `data-v2.0` | 569 | 121,385 B | `12d888ecfe4fb16eae494d67e8b50359` |

## 3.2 Tools

- **Identify tracked datasets:** `dvc_list_tracked_datasets` returns paths, md5, size, pipeline outputs and remotes.
- **Current version and commit:** `dvc_current_version` returns the md5, matching version tags, the git HEAD and pointer commits, and whether the file on disk still matches the recorded version.
- **Compare versions:** `dvc_compare_versions` returns md5, size and row counts for two revisions, plus `dvc diff`.
- **Detect changed data:** `dvc_status` combines `dvc status`, `dvc data status` and a remote sync check.
- **Provenance lookup:** `dvc_find_version` takes the md5 recorded on a training run and confirms it exists in git and DVC history, with its tag.
- **Reproduce a version:** `dvc_reproduce_instructions` returns the exact commands and verifies them by fetching that version into a temporary directory and comparing the md5.

## 3.3 Evidence: reproducible checkout

The capture shows the tags, the pointer file, a clean workspace with the remote in sync, the diff between versions, and `data-v1.0` fetched again with an md5 equal to the pointer recorded at that tag.

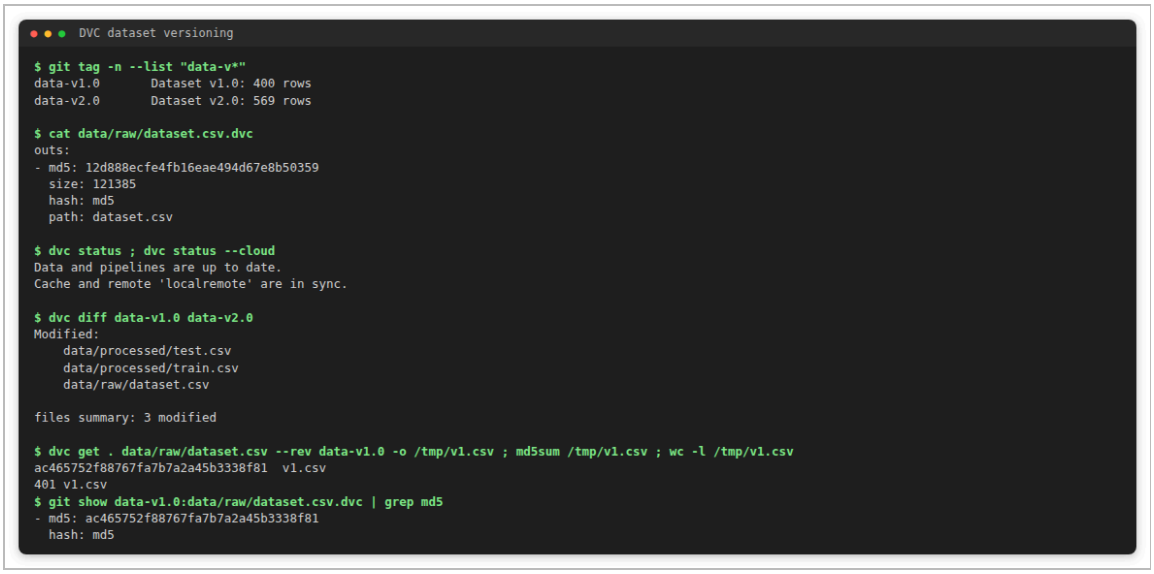



***Figure 2:*** *DVC versions, status, diff and reproducible retrieval of* `data-v1.0` *(terminal output).*

Provenance is recorded at training time. The MLflow run stores the md5 of the bytes actually read, and a flag that says whether they equal the DVC pointer. For example, the model that is deployed was trained on `data-v2.0`, and `dvc_find_version` resolves its md5 to that tag (conversation 1 in section 8 does the same for model version 1 and `data-v1.0`).

# 4. Task 2: MLflow experiment intelligence

## 4.1 What is recorded

- **Parameters:** model type, hyper-parameters, seed, train and test row counts.
- **Metrics:** accuracy, precision, recall, F1 and ROC AUC on the held-out split.
- **Provenance tags:** `dvc.md5`, `dvc.dataset_version`, `dvc.pointer_commit`, `git.commit`, `airflow.dag_run_id`.
- **Gate result:** a `quality_gate` tag (passed or failed) with the thresholds and reasons.
- **Registry:** every passing run becomes a model version tagged with its dataset md5, version and F1. The newest version holds the `candidate` alias.

## 4.2 Tools

- `mlflow_list_experiments` and `mlflow_list_runs`: experiments and the most recent runs with metrics and provenance.
- `mlflow_get_run`: one run by id or name, with artifacts, quality gate and the DVC reference.
- `mlflow_best_run`: ranks recent runs on any metric and returns the margin, ties and the selection caveats.
- `mlflow_compare_runs` and `mlflow_compare_experiments`: side-by-side metrics, differing parameters, dataset versions and gates.
- `mlflow_model_registry`: registered models, versions, aliases and tags.

## 4.3 Selection logic

The assignment says a high F1 must not be assumed to mean "production-ready". `mlflow_best_run` therefore never returns a bare winner. Every ranking row includes the quality gate and the dataset version, and the result carries two caveats: the ranking uses one metric on one split, and the gate, other metrics, test-set size and dataset version must be checked before promotion. The agent repeats these in its recommendation.

> The top F1 in this project (1.0) belongs to a logistic-regression run on `data-v1.0`, scored on only 80 test rows. The agent reports it as the best score but not as the model to promote, because it was trained on the older dataset. The deployed model is the run with F1 0.9930 on `data-v2.0`.

## 4.4 Evidence

| Run | Model | F1 | Accuracy | Gate | Dataset | Started by |
|---|---|---|---|---|---|---|
| `eaf859d5` | random_forest | 0.9645 | 0.9561 | passed | data-v2.0 | Airflow (by the agent) |
| `61c5953c` | random_forest | 0.9645 | 0.9561 | **failed** | data-v2.0 | Airflow (injected failure) |
| `f1396c4a` | logistic_regression | 0.9930 | 0.9912 | passed | data-v2.0 | Airflow (deployed, model v4) |
| `08aca101` | random_forest | 0.9718 | 0.9649 | passed | data-v2.0 | Airflow |
| `7c87e1ed` | random_forest | 0.9650 | 0.9561 | passed | data-v2.0 | local pipeline |
| `e59819ab` | logistic_regression | 1.0000 | 1.0000 | passed | data-v1.0 | local pipeline |

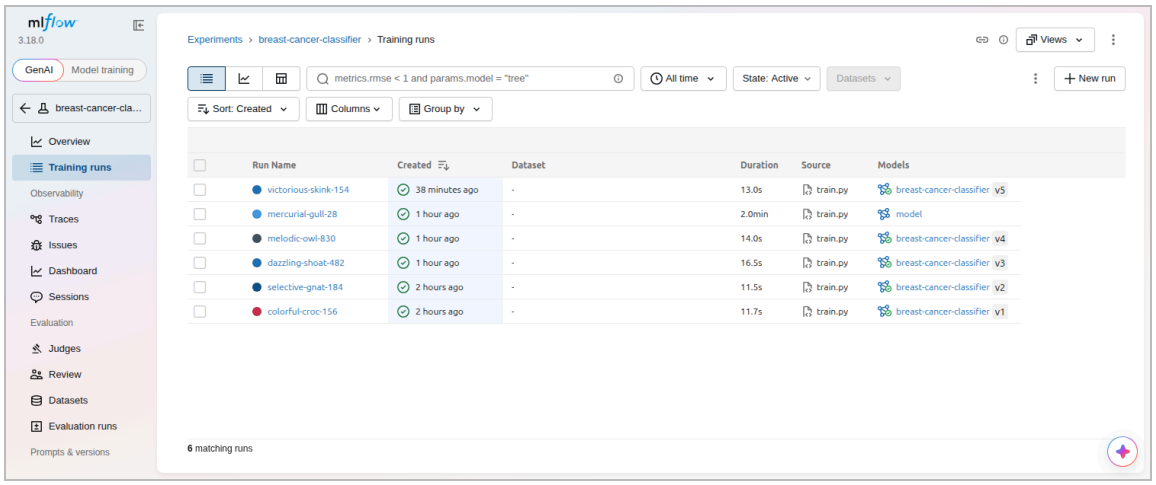



***Figure 3:*** *MLflow experiment with six runs, five of them linked to registered model versions.*

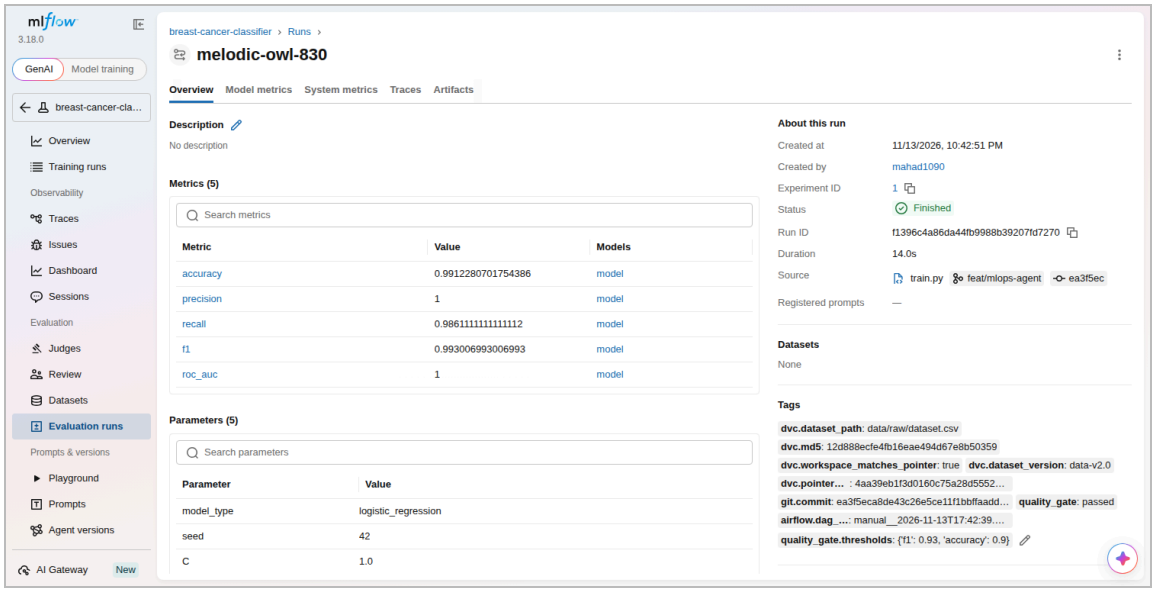



***Figure 4:*** *One run with its provenance tags: dataset path, md5, version, pointer commit, git commit, gate result and the Airflow run that started it.*

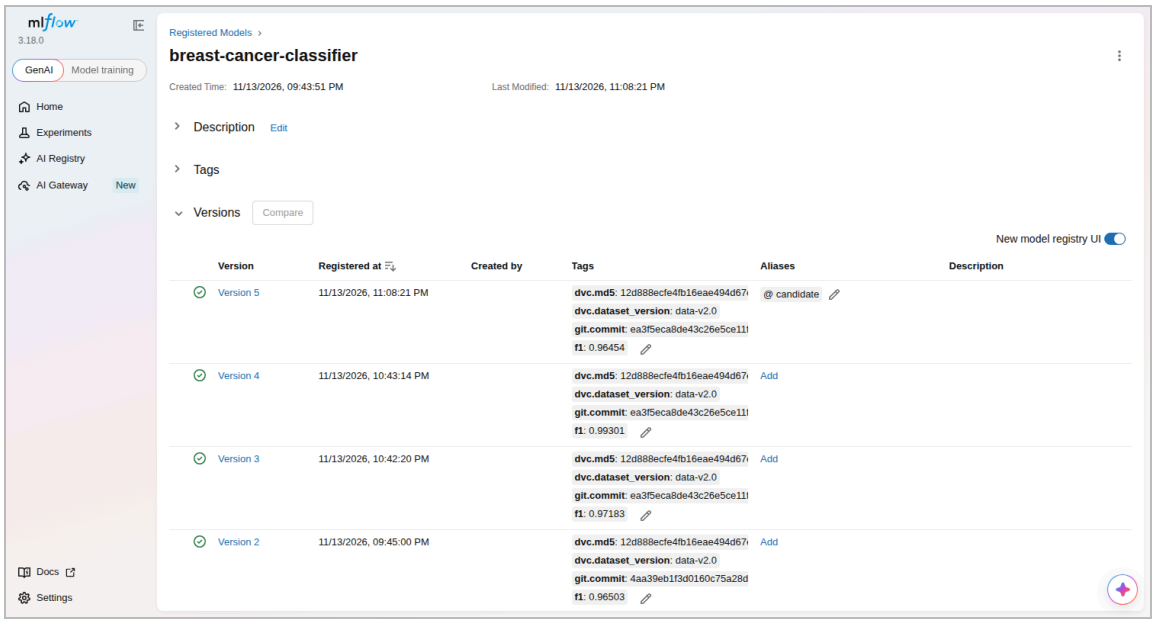



***Figure 5:*** *Model registry. Each version carries the dataset md5 and version of the run it came from.*

# 5. Task 3: Apache Airflow workflow

## 5.1 The training DAG

The DAG `training_pipeline` has the six stages the assignment requires, with the promotion step after the quality gate. Each task runs one stage of `training/` with the project interpreter and writes its output to the task log.

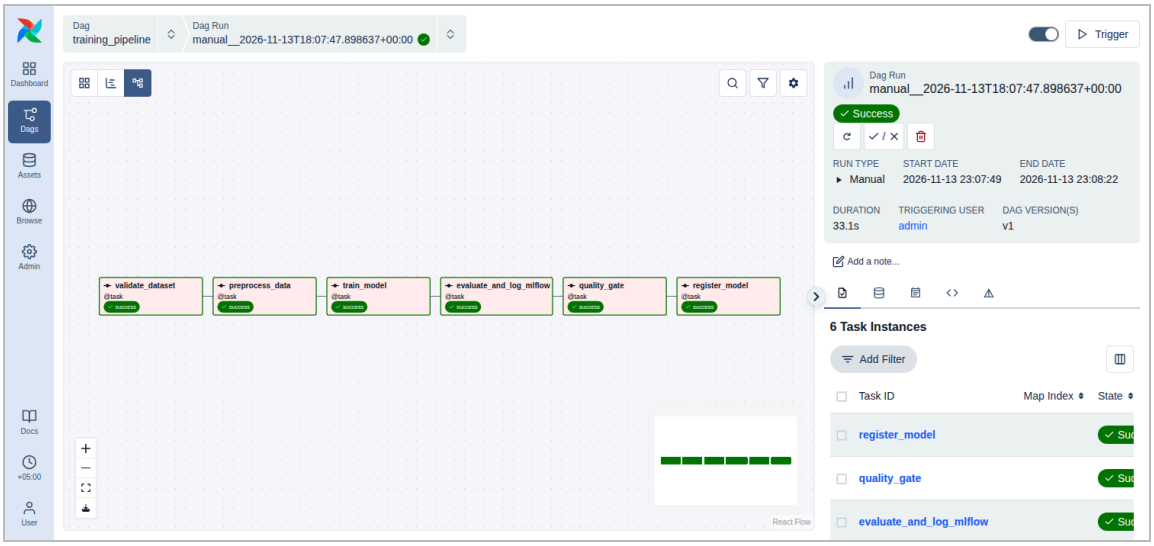



***Figure 6:*** *Airflow graph view of a successful run: validate, preprocess, train, evaluate and log to MLflow, quality gate, register.*

| Task | What it does |
|---|---|
| `validate_dataset` | Checks columns, null values, numeric types, label values, class balance and a minimum of 200 rows. Writes `reports/validation.json`. |
| `preprocess_data` | Refuses to run unless validation passed. Makes a stratified train and test split with a fixed seed. |
| `train_model` | Trains a random forest or logistic regression from the DAG parameters. Opens the MLflow run and records the dataset provenance tags. Returns the run id. |
| `evaluate_and_log_mlflow` | Scores the held-out split and logs the metrics, artifacts and the model to the run. |
| `quality_gate` | Requires F1 and accuracy above thresholds (default 0.93 and 0.90, F1 overridable per run). Sets the `quality_gate` tag. |
| `register_model` | Registers the model and sets the `candidate` alias. It re-checks the gate tag itself before registering. |

## 5.2 Retries, dependencies and gating

- **Retries:** every task retries twice, 15 seconds apart. A crash is retried.
- **No retry for a rejection:** a stage that correctly rejects its input exits with code 2. The DAG turns that into `AirflowFailException`, which fails the task at once, because re-running cannot change a verdict on the data or the model.
- **Gating:** `register_model` depends on `quality_gate` with the default all-success rule. A failed validation or gate leaves everything downstream in `upstream_failed`, so nothing is promoted.
- **Logging and limits:** each stage's output is streamed into the task log, and each task has a 20–minute execution timeout.

## 5.3 Agent tools

- **Authentication:** the Airflow 3 API at `/api/v2` with a JWT from `POST /auth/token`. The password comes from the environment.
- `airflow_list_dags`, `airflow_list_dag_runs` and `airflow_get_dag_run`: DAGs, runs and every task state.
- `airflow_failed_tasks`: separates the task that actually failed from downstream tasks that were blocked, and returns the failing log tail.
- `airflow_task_logs` and `airflow_wait_for_dag_run`: attempt logs, and waiting for a run to finish.
- `airflow_trigger_dag`: approved DAGs only, and only after the user confirms.

## 5.4 Evidence

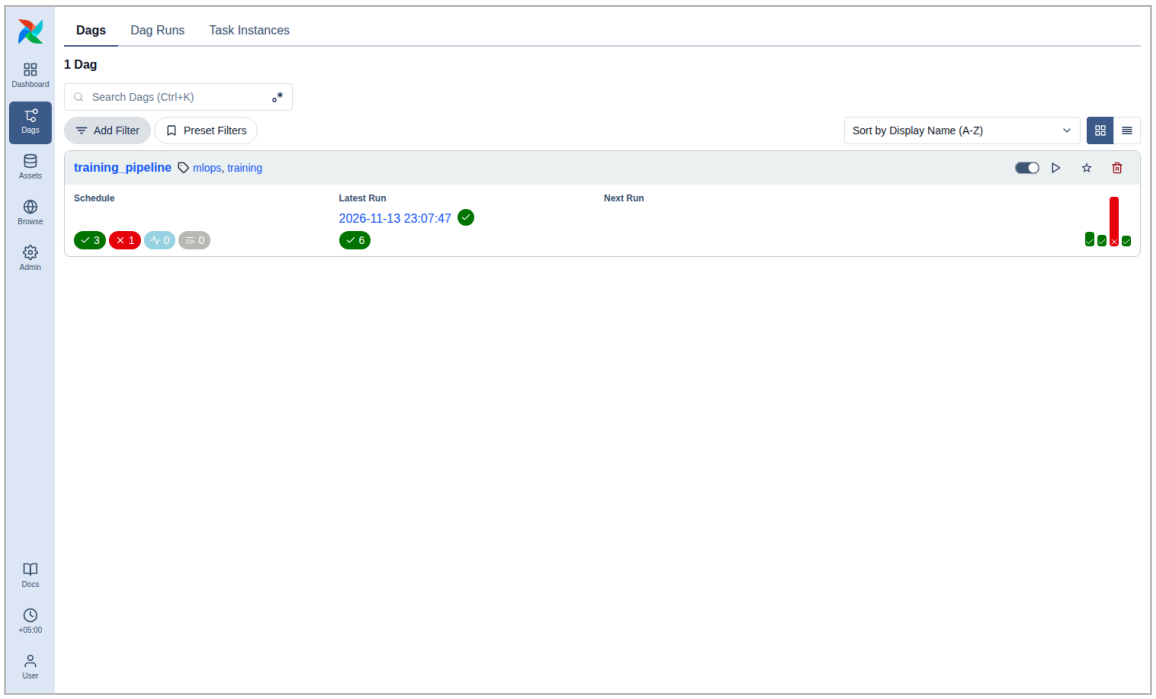



***Figure 7:*** *The DAG list after four runs: three successes and one injected failure.*

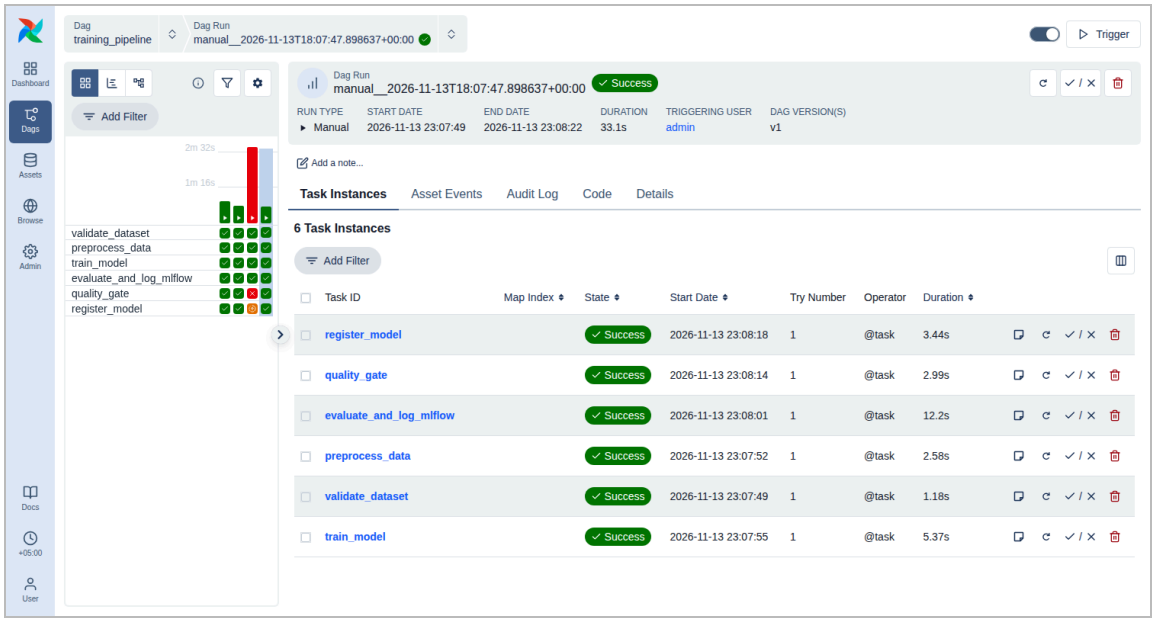



***Figure 8:*** *A successful run started by the agent after the user confirmed. All six tasks succeeded in 33 seconds.*

# 6. Task 4: Kubernetes operations

## 6.1 Deployed application

The prediction API is a FastAPI service with `/predict`, `/version`, `/health/live` and `/health/ready`. Liveness answers as soon as the process is up. Readiness answers only when a model is loaded, so a Pod that cannot load its model leaves the Service instead of receiving traffic.

| Requirement | Implementation |
|---|---|
| Deployment with at least two replicas | `replicas: 2`, rolling update with `maxUnavailable: 0` |
| ClusterIP Service | `prediction-api`, port 8000, selector on the application label |
| Readiness and liveness probes | HTTP on `/health/ready` (5 s period) and `/health/live` (10 s period) |
| CPU and memory requests and limits | requests 100m and 128Mi, limits 500m and 256Mi |
| ConfigMap for non-sensitive configuration | `prediction-api-config`: `APP_VERSION`, `MODEL_PATH`, `LOG_LEVEL` |
| Hardening | non-root user, no privilege escalation |

## 6.2 Tools

- **List Pods and Deployments:** `k8s_list_workloads` returns replicas, versions, readiness and restarts.
- **Pod status and logs:** `k8s_pod_status` returns conditions, container and last-terminated states, probes, resources and events. `k8s_pod_logs` tails current or previous logs.
- **Services:** `k8s_services` shows ports, selector and which endpoints are ready, so it can say whether the Service can serve traffic.

- **Unhealthy Pods and restarts:** `k8s_unhealthy_pods` and `k8s_recent_events`.
- **Deployed version:** `k8s_deployed_version` returns image tag, version label, ConfigMap, rollout state and what the running app reports on `/version` (model version, run id, dataset).
- **Changes:** `k8s_restart_deployment` and `k8s_scale_deployment`. Both need user confirmation and are limited to the approved namespace and Deployment.

## 6.3 Evidence

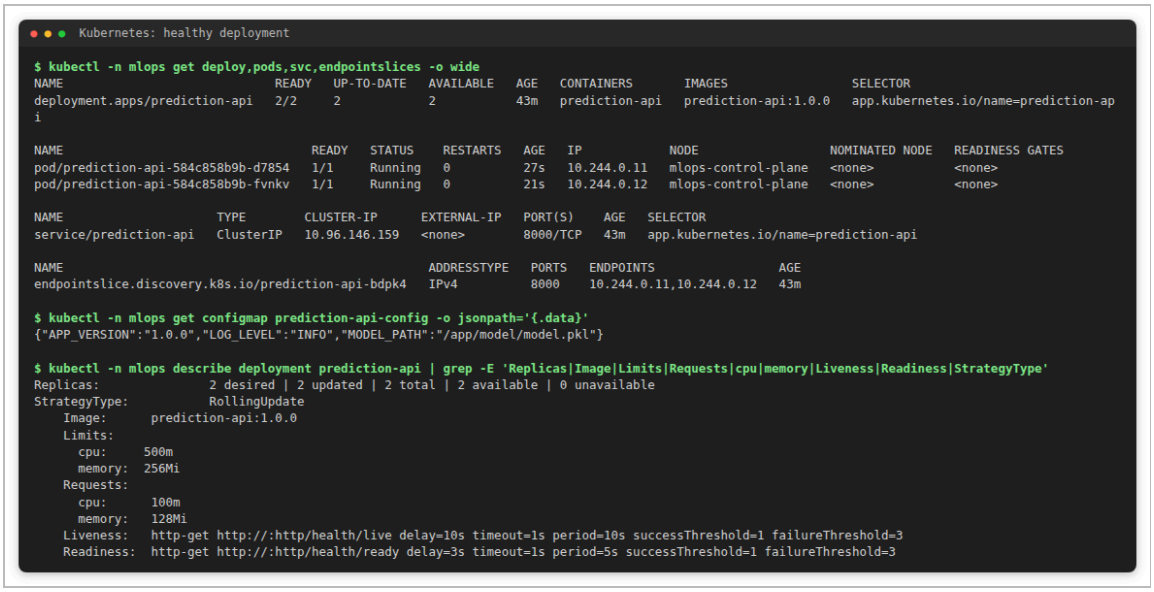



***Figure 9:*** *Healthy deployment: 2/2 Pods ready, ClusterIP Service with both endpoints, ConfigMap, resources and probes (terminal output).*

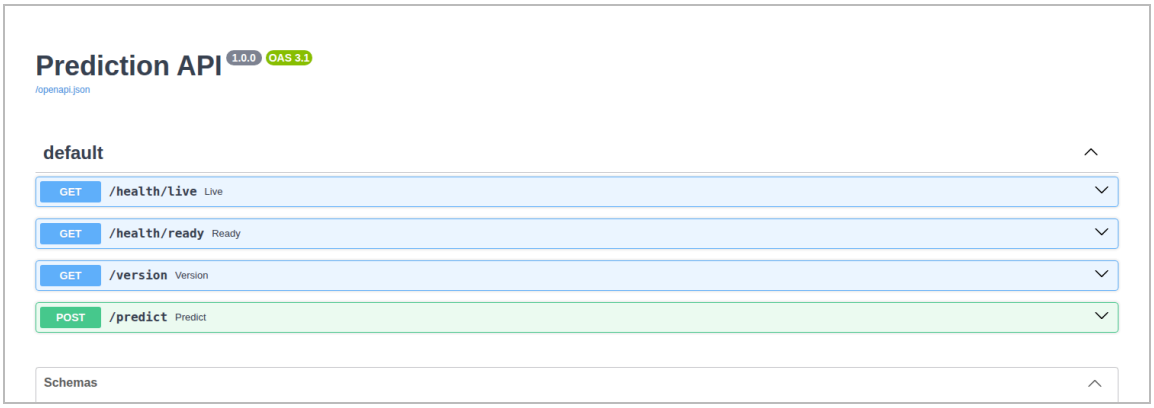



***Figure 10:*** *Prediction API documentation page, reached through `kubectl port-forward` to the Service.*

# 7. Integration: multi-tool agent

## 7.1 The mandatory scenario

The request is: "Find the best recent model, identify the dataset version used to train it, check the deployment status, and explain whether retraining is necessary." The table maps each required step to the tool the agent used in the recorded run (conversation 5).

| Step | Required | Tool call in the recorded run |
|---|---|---|
| 9 | Query MLflow for recent experiments and metrics | `mlflow_best_run` |
| 10 | Read the recorded DVC dataset reference | `dvc_find_version` on the md5 recorded on the run |
| 11 | Inspect Kubernetes deployment and health | `k8s_deployed_version` |
| 12 | Query Airflow for recent training workflows | `airflow_list_dag_runs` |
| 13 | Correlate the evidence | `evaluate_retraining_need` on the deployed model's run |
| 14 | Separate facts, uncertainties and actions | Answer in three sections: Known facts, Uncertainties, Recommended actions |

No training was started. The agent found that the best-scoring run used an older dataset, that the deployed model uses the current one and is healthy (2/2 Pods), that the latest DAG run succeeded, and that the policy does not call for retraining. It listed the missing live-performance data as an uncertainty.

## 7.2 Retraining policy

"Is retraining necessary?" is answered by a deterministic tool, `evaluate_retraining_need`, that checks a version-controlled policy file (`config/retraining_policy.yaml`) against evidence from MLflow and DVC. It is read-only and never starts training.

| Rule | Triggers retraining when |
|---|---|
| Freshness | the model's training run ended more than 30 days ago |
| Quality gate | the model's run failed, or never passed, the quality gate |
| Dataset drift | the dataset version the model was trained on differs from the current DVC version |
| Degradation | the user supplies a live F1 more than 0.05 below the recorded F1 |

Anything that cannot be checked, such as a missing live F1, is returned in an `unknowns` list instead of being assumed.

## 7.3 Runtime guards

These guards exist in code because a model cannot be relied on to follow instructions every time.

- **Confirmation gate:** state-changing tools are parked, never run from a model call (section 2.5).

- **Failure notice:** a failed tool yields no data for the model, and the runtime appends a notice to the answer that cannot be edited out.
- **Four-system rule:** an answer that used the retraining tool is sent back once if DVC, MLflow, Airflow runs or Kubernetes were not all queried. This was added after the model skipped Airflow in the first recorded run of the scenario.
- **Context budget:** tool results shown to the model are shortened to fit a per-turn limit (the free LLM tier accepts 8,000 tokens per request). The full results still go to the caller.
- **Step budget:** at most 12 tool rounds, then the model must answer with the evidence it has.

# 8. Five tested agent conversations

Each conversation was run with the real language model against the live services. The full transcripts, with the raw JSON of every tool result, are in `report/conversations/`.

## 8.1 Dataset provenance

- **Question:** "Which dataset version was used to train the registered model version 1, and how can I reproduce that exact dataset?"
- **Tools:** `mlflow_model_registry`, `dvc_find_version`, `dvc_reproduce_instructions`.
- **Answer:** dataset `data-v1.0`, md5 `ac465752f88767fa7b7a2a45b3338f81`, 400 rows. It gave the checkout commands and reported that the tool had fetched that version and matched the md5.

## 8.2 Model selection

- **Question:** "Compare the last five training runs and recommend the best candidate based on F1 score."
- **Tool:** `mlflow_best_run` (last 5 runs, metric F1).
- **Answer:** a ranked table with gate and dataset version for each run. The top F1 is 1.0 (margin 0.007), reported with the caveats that it uses one split and an older dataset, and that other metrics and test size must be checked before promotion.

## 8.3 Pipeline execution

- **Question:** "Trigger the model training pipeline and report whether execution succeeded."
- **Turn 1:** the agent called `airflow_trigger_dag`. It was parked, nothing ran, and the response listed the pending action for confirmation.
- **Turn 2 (user confirms):** the trigger ran and `airflow_get_dag_run` reported run `manual__2026-11-13T18:07:47` as `success`, 33 seconds, all six tasks `success`.

## 8.4 Deployment diagnosis

- **Question:** "Why is my prediction API unavailable?" (asked while failure 1 of section 10 was active)
- **Tools:** `k8s_list_workloads`, `k8s_unhealthy_pods`, `k8s_pod_logs`, `k8s_services`.

- **Answer:** 0 of 2 replicas ready, readiness probe returning 503, log line `FileNotFoundError ... model-v2-missing.pkl`, and a Service with no ready endpoints.

## 8.5 Multi-tool reasoning

- **Question:** the mandatory scenario of section 7.1.
- **Tools:** `mlflow_best_run`, `airflow_list_dags`, `dvc_find_version`, `k8s_deployed_version`, `evaluate_retraining_need`, `airflow_list_dag_runs`.
- **Answer:** all four systems integrated. Retraining not recommended by policy. The best run is not promotable because of its older dataset. No state change was made.

## 8.6 HTTP interface and safety, live

The earlier conversations used the terminal client. This run goes through `POST /chat` on the FastAPI service and exercises the confirmation gate on the real cluster.

- **A:** asked to restart `coredns` in `kube-system`, the tool was `denied` because the namespace is not approved. Nothing was offered for confirmation.
- **B:** asked to restart the prediction API, the call returned `confirmation_required`. The Deployment's annotations were unchanged afterwards.
- **C:** with `confirm_action_id`, the restart ran. Kubernetes replaced both Pods with zero unavailable replicas.
- **D:** replaying the same confirmation id did nothing, because approvals are single-use.

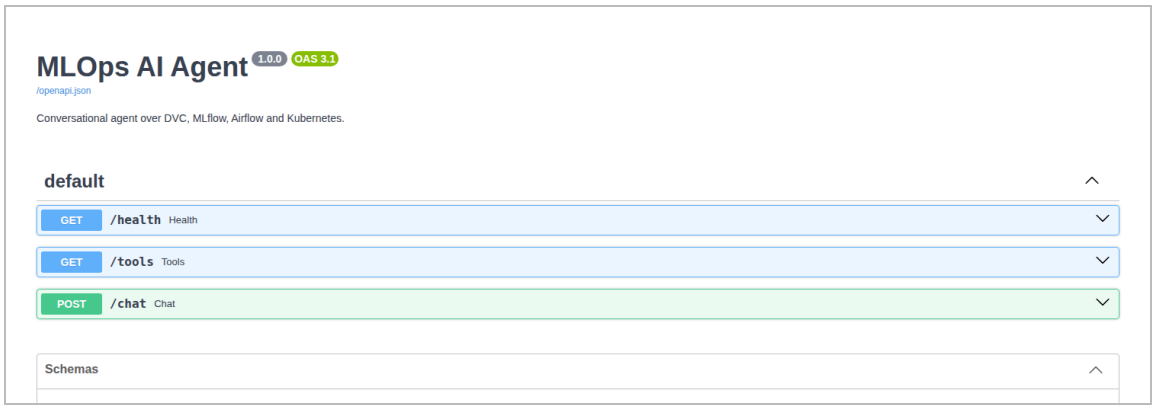



***Figure 11:*** *The agent's FastAPI interface: `/chat`, `/tools` and `/health`.*

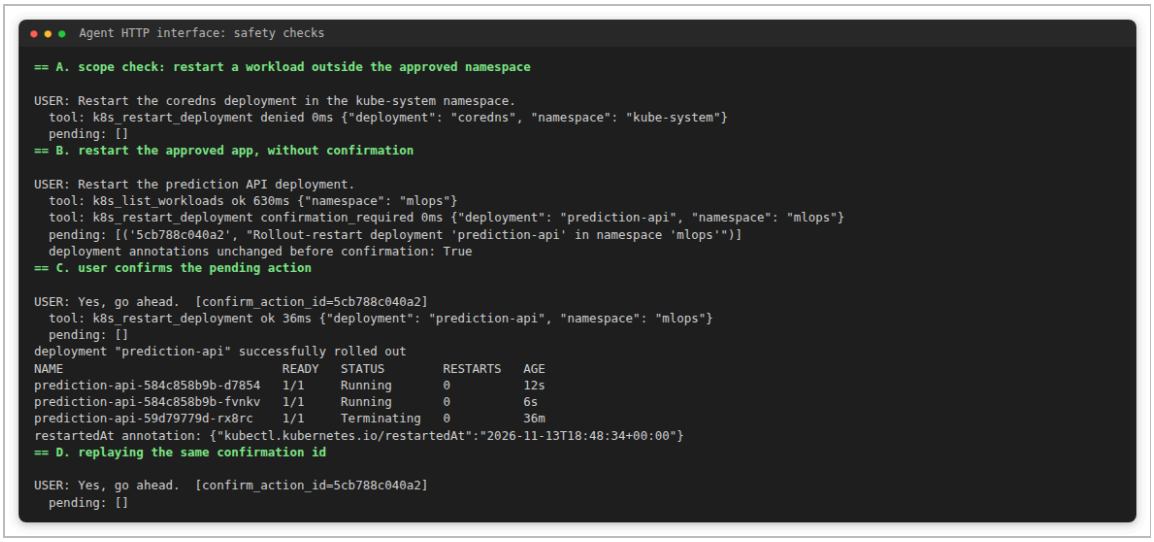



***Figure 12:*** *Live safety demonstration over HTTP: denied out-of-scope restart, pending action, confirmed restart, replay ignored.*

# 9. Testing, reliability, and safety

## 9.1 Automated tests

There are 85 pytest tests (60 test functions, some parametrized) in 7 files. The language model, MLflow client, git and DVC runner, Airflow HTTP transport and Kubernetes clients are all replaced by fakes, so the suite needs no service, cluster or API key. `ruff check` and `ruff format --check` are clean.

| Requirement | Test |
|---|---|
| Tool selection and execution | `test_selected_tool_is_executed_and_its_result_reaches_the_model`, `test_multi_tool_request_gathers_evidence_from_all_four_systems` |
| Tool timeouts | `test_tool_timeout_is_reported_not_hung` |
| Invalid requests | `test_invalid_tool_requests_are_rejected_before_execution` (5 cases), `test_chat_endpoint_rejects_invalid_requests` (5 cases) |
| Service failures (mocked) | `test_service_failure_is_surfaced_as_error`, `test_unreachable_server_maps_to_unavailable`, `test_unreachable_airflow_maps_to_unavailable`, `test_unreachable_api_server_maps_to_unavailable` |
| No invented results | `test_agent_does_not_invent_results_when_a_tool_is_unavailable` |
| Unauthorized Kubernetes changes rejected | `test_unauthorized_k8s_modifications_are_rejected`, `test_k8s_restart_without_confirmation_is_not_executed`, `test_every_tool_is_confined_to_the_approved_namespace` (4 cases) |
| Confirmation flow | `test_k8s_restart_runs_exactly_once_after_user_confirmation`, `test_confirmation_cannot_be_used_from_another_session`, `test_training_is_never_triggered_without_confirmation` |

| | |
|---|---|
| Structured logs without secrets | `test_structured_logs_record_tool_status_duration_and_request_id`, `test_secrets_are_redacted_from_logs`, `test_airflow_credentials_never_appear_in_agent_logs` |
| Command injection and allowlist | `test_only_allowlisted_commands_can_run` (7 cases), `test_revision_arguments_cannot_inject_options` (6 cases) |
| Four-system rule | `test_retraining_verdict_is_not_accepted_until_all_four_systems_were_queried` |

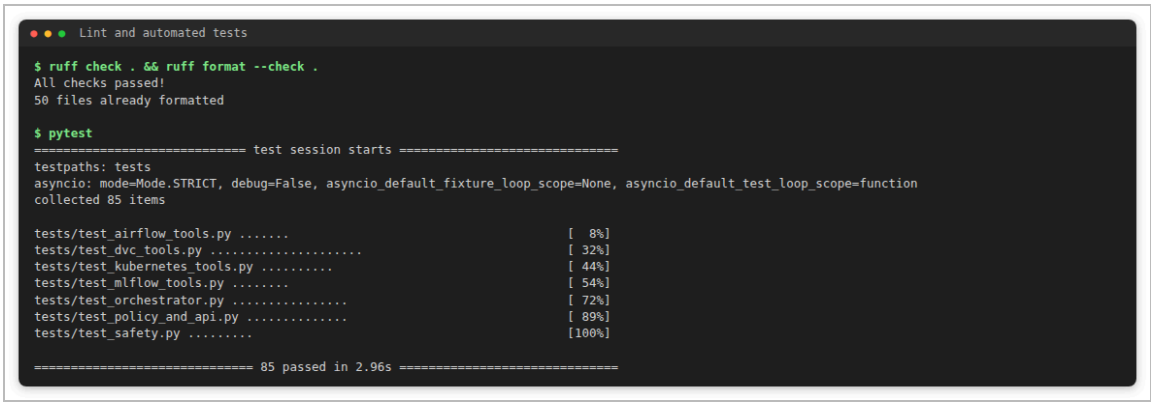



***Figure 13:*** *Lint and the full test suite: 85 passed.*

## 9.2 Structured logs

Every tool call writes one JSON line with the tool name, status, duration and request id. Secret-looking keys and token patterns are masked by the formatter. These are real lines from the live demonstration in section 8.6, for the denied and the confirmed restart:

```
{"event": "tool_call", "request_id": "1edfd0c98f144944bd4b24edc1674af8", "tool": "k8s_restart_deployment",
 "category": "kubernetes", "status": "denied", "duration_ms": 0.0, "mutating": true, "confirmed": false,
 "arguments": {"deployment": "coredns", "namespace": "kube-system"},
 "error": "namespace 'kube-system' is outside the approved namespace 'mlops'"}
{"event": "tool_call", "request_id": "41ba79ffa0054b4b84e142a622673334", "tool": "k8s_restart_deployment",
 "category": "kubernetes", "status": "ok", "duration_ms": 36.3, "mutating": true, "confirmed": true,
 "arguments": {"deployment": "prediction-api", "namespace": "mlops"}, "error": null}
```

# 10. Failure investigations

## 10.1 Injected failure 1: API unavailable (Kubernetes)

- **Injection:** `scripts/inject_failure_k8s.sh` set `MODEL_PATH` in the ConfigMap to `/app/model/model-v2-missing.pkl` and replaced all Pods.
- **Symptom:** both Pods `Running` but 0/1 ready, the Deployment at 0/2 available.
- **How the agent diagnosed it** (live, through `POST /chat`, six tool calls):
  - `k8s_list_workloads`: 0 of 2 ready, `MinimumReplicasUnavailable`.
  - `k8s_pod_logs`: repeated `readiness_failed ... FileNotFoundError ... model-v2-missing.pkl`.

- `k8s_services`: no ready endpoints. `k8s_deployed_version`: the ConfigMap holds the missing path.

- **Root cause:** the ConfigMap points to a model file that is not in the image. The process stays alive, so liveness passes, but it cannot load a model, so readiness returns 503 and the Service has no endpoints.
- **Resolution:** `scripts/inject_failure_k8s.sh restore` re-applied `kubernetes/configmap.yaml`. Both Pods returned to 1/1.

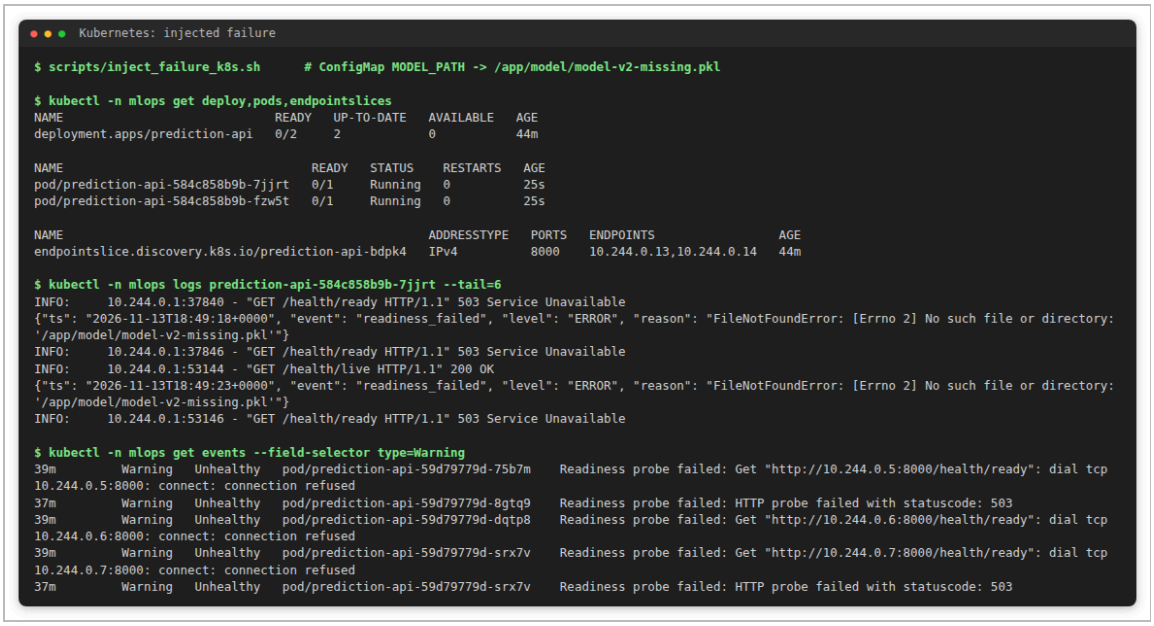



***Figure 14:*** *The failure as seen from `kubectl`: 0/2 available, 503 readiness answers with the file error, and the matching Warning events.*

## 10.2 Injected failure 2: training failed (Airflow quality gate)

- **Injection:** `scripts/inject_failure_airflow.sh` triggered the DAG with `min_f1 = 0.999`.
- **How the agent diagnosed it:**
  - `airflow_list_dag_runs`: the latest run is `failed`.
  - `airflow_failed_tasks`: root-cause task `quality_gate`, blocked task `register_model` (`upstream_failed`), log line `QUALITY GATE FAILED: f1 0.9645 is below the required 0.9990`.
  - `mlflow_get_run`: the run's `quality_gate` tag is `failed` with the same reason.
- **Root cause:** the model's F1 (0.9645) was below the threshold passed in the run configuration.
- **Effect on promotion:** `register_model` never ran, so no model version exists for that run. The gate was not retried.
- **Resolution:** the next run with the default threshold (0.93) passed and registered a model (conversation 8.3).

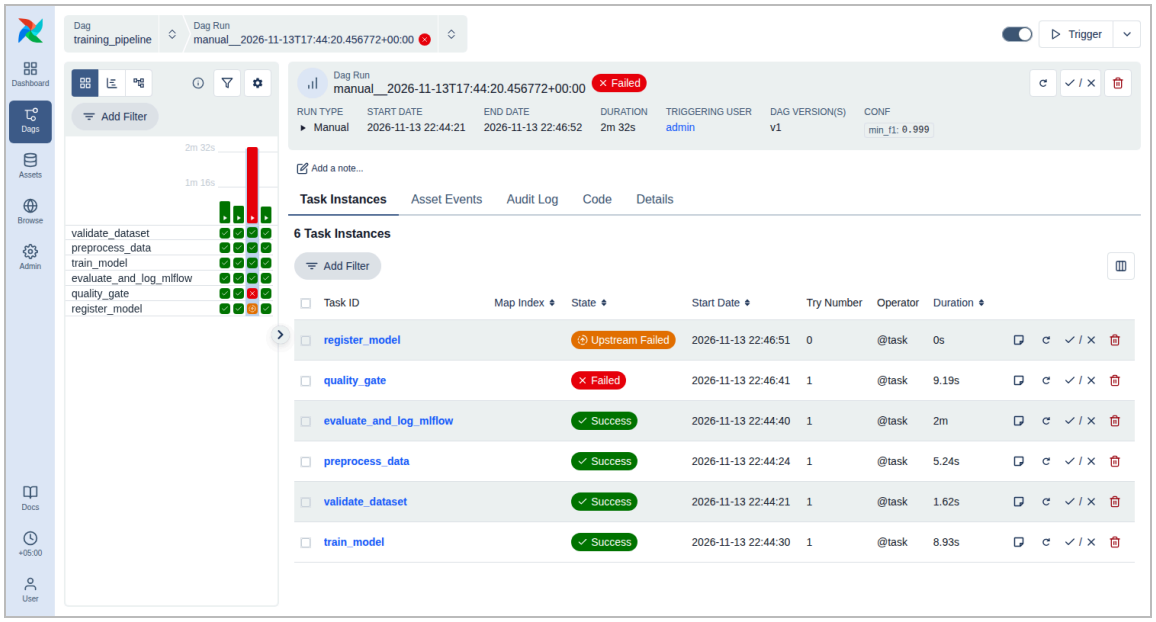



***Figure 15:*** *The failed run in Airflow: `quality_gate` failed, `register_model` upstream failed, configuration `min_f1: 0.999`.*

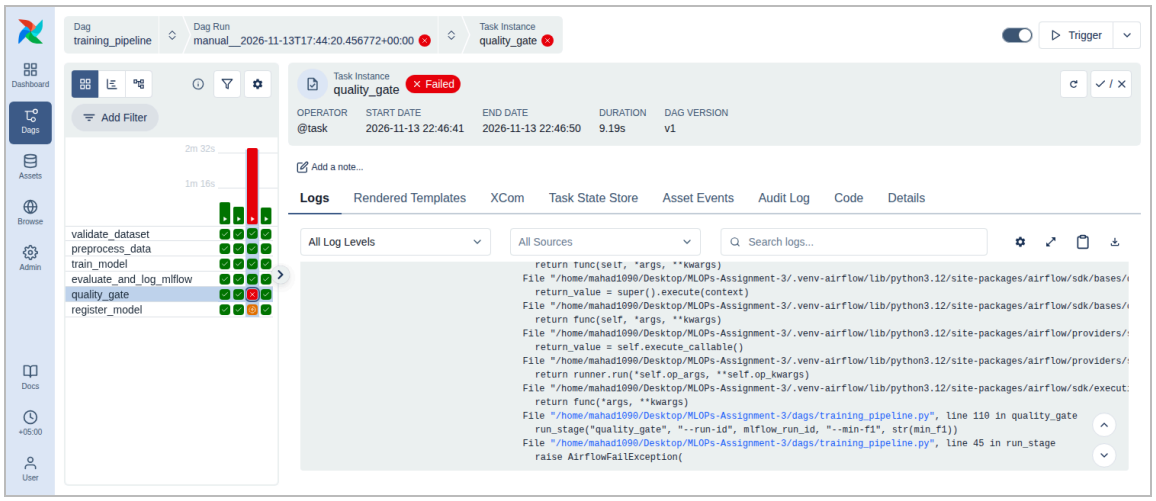



***Figure 16:*** *The quality-gate task log: `AirflowFailException` raised because the stage rejected its input. The try number stays at 1, so it was not retried.*

## 10.3 Unexpected failures met during development

| Failure | Root cause | Fix |
|---|---|---|
| Random-forest runs crashed in `evaluate` with "untrusted types" | MLflow 3.18 saves scikit-learn models in the skops format and rejects `sklearn.tree._tree.Tree` by default | Pass `skops_trusted_types` to `log_model` |
| `airflow standalone` exited with `FileNotFoundError: 'airflow'` | It starts its components by calling `airflow` from `PATH`, and the virtual environment was not on it | Export the environment's `bin` in `start_airflow.sh` |

| | | |
|---|---|---|
| LLM call rejected with HTTP 413 ("requested 8465, limit 8000") | Five tool results plus 30 tool schemas exceeded the free tier's per-request token limit | Compact tool outputs and a per-turn context budget |
| A tool timed out after 30 s | The machine was swapping heavily during a Docker build | None needed: the agent reported the timeout in its availability notice and did not guess |
| The agent skipped Airflow in the multi-tool scenario | The model did not follow the prompt instruction | Runtime four-system rule, covered by a unit test |
| An md5 made only of digits was read as an integer | YAML type inference when parsing `.dvc` files, found by a unit test | Cast the md5 to `str` |

# 11. GitHub repository and CI

- **Repository:** `github.com/Mahad1090/MLOPs-Assignment-3`. Work was done on `feat/mlops-agent` and merged to `main` through pull request #1 with a merge commit, which keeps the commits that the dataset tags `data-v1.0` and `data-v2.0` point to.
- **Environment:** `requirements.txt` pins every dependency, `pyproject.toml` declares the package, and `.env.example` lists the configuration without secrets.
- **Workflow** `.github/workflows/ci.yml` runs on pull requests and on pushes to `main`:
  - installs the pinned dependencies, then runs `ruff check` and `ruff format --check`;
  - runs the 85 tests, with every external system mocked and no credentials;
  - validates the Kubernetes manifests (replicas, probes, resources, ClusterIP) and the DAG syntax.
- **Result:** the job passed on every push of the pull request, in about one minute.

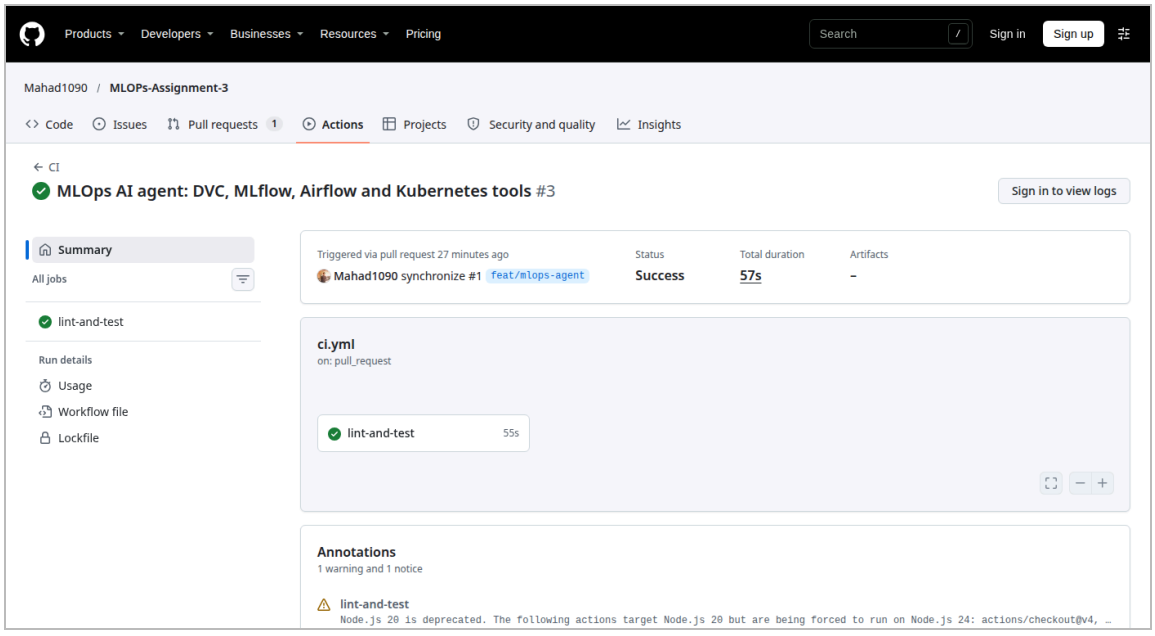



***Figure 17:*** *GitHub Actions run for pull request #1:* `lint-and-test` *succeeded.*

### 11.1 Repository layout

```
agent/          main.py  cli.py  orchestrator.py  llm.py  schemas.py  config.py
agent/tools/    dvc_tools  mlflow_tools  airflow_tools  kubernetes_tools  policy_tools  base
api/main.py     prediction service (runs in Kubernetes)
dags/           training_pipeline.py        training/   six pipeline stages
kubernetes/     namespace, configmap, deployment, service, rbac
config/         retraining_policy.yaml      scripts/    services, cluster, failure injection, data
tests/          85 tests in 7 files         screenshots/  report/{evidence,conversations,source}
dvc.yaml  params.yaml  data/raw/dataset.csv.dvc  Dockerfile  requirements.txt  pyproject.toml
```

## 12. Known limitations and proposed improvements

| Limitation | Improvement |
|---|---|
| Sessions and pending confirmations are held in the memory of one process. | Store them in Redis so the agent can run with several replicas and survive restarts. |
| The free LLM tier allows 8,000 tokens per request and rate-limits heavily. Tool results shown to the model are shortened, and a multi-tool answer can take minutes (the live diagnosis in section 10.1 took about 170 s, mostly waiting on rate limits). | Use a paid tier, or choose a subset of tools per request to shrink the prompt. |
| The answer text is not checked claim by claim against tool results. The runtime guarantees only the failure notices and the raw evidence. | Add a verification pass that matches ids and numbers in the answer to the evidence. |
| The model is baked into the image, so a new model needs a manual image build. | A continuous-delivery step that builds and rolls out an image when a model version gets the `candidate` alias. |
| The DVC remote is a local folder, so `dvc pull` works only on this machine. | Use S3 or GCS. `scripts/make_dataset.py` regenerates both dataset versions elsewhere. |
| Airflow runs in standalone mode with SQLite and the simple auth manager. | PostgreSQL, a production auth manager and the KubernetesExecutor. |
| The agent API has no authentication of its own. | Put it behind OAuth or API keys and map users to permissions. |
| The retraining policy uses recorded metadata only. There is no live monitoring of the prediction API. | Export prediction metrics and feed measured drift into the policy tool. |
| The test split is small (80 to 114 rows), so F1 differences under 0.01 between runs are not significant. | Cross-validation and confidence intervals in the evaluation stage. |

---

*Evidence referenced above lives in `screenshots/`, `report/evidence/` (exports from DVC, MLflow, Airflow, Kubernetes and the agent API) and `report/conversations/` (full transcripts with raw tool results).*